

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	02 July 2026
Team ID	SWTID-2026-4978
Project Name	Strategic Product Placement Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Connection & Parsing	- Connect natively to raw sales dataset logs without schema alteration. - Map row positioning dimensions (Aisle, End-cap, Front) cleanly into variables.
FR-2	Spatial Volume Analysis	- Construct an interactive Treemap rendering to group total sales volumes. - Allow quick visual identification of highest-producing retail floor sectors.
FR-3	Demographic Segmentation	- Isolate purchasing habits by customer segments via donut chart visuals. - Integrate global dashboard action filters to update path selections.
FR-4	Market Benchmark Mapping	- Display side-by-side bar metrics comparing average prices vs. competitors. - Render visual indicators highlighting pricing over-indexing markers.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard interface must feature clear tooltips, legends, and a zero-code interactive design suitable for non-technical store operational managers.
NFR-2	Security	The workbook package must operate locally or within secure server nodes to prevent unauthorized exposure of internal pricing margins and target sales metrics.
NFR-3	Reliability	Cross-filtering actions and sheet interactions must execute consistently without fracturing calculated fields or rendering null graphical values.
NFR-4	Performance	All chart animations, dimension recalculations, and dashboard global filter actions must render cleanly in less than 2 seconds upon interaction.

FR No.	Non-Functional Requirement	Description
NFR-5	Availability	The compiled file must be accessible via any standard browser via Tableau Public/Server to accommodate category planners working remotely.
NFR-6	Scalability	The underlying engine logic must support expanding rows to accommodate higher monthly transaction logging volumes without slow execution drops.